ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **Exploring Substances: Acidic, Basic, and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** indicators	e.g. litmus, turmeric, China rose
2	**(a)** Soap solution	feels soapy and turns red litmus blue
3	**(d)** tartaric acid	gives sour taste
4	**(b)** Baking soda paste	a base neutralises the acid
5	**(c)** neutralisation	gives salt and water
6	**(d)** lime (slaked lime)	a base
7	**(c)** Onion	its smell is lost in bases
8	**(a)** basic	mildly basic
9	**(b)** acidic	acid test
10	**(d)** damages buildings and water bodies	acidic precipitation

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substances that show different colours in acidic and basic solutions (1); litmus and turmeric (1).	1+1
12	Blue litmus turns red in an acid (2).	2
13	Acids taste sour and turn blue litmus red (1); bases taste bitter and feel soapy (1).	1+1
14	Reaction of an acid with a base giving salt and water (1); antacid with stomach acid (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Put drops on blue and red litmus (1); lemon turns blue to red, soap turns red to blue (1); sugar no change (1).	1+1+1
16	Crush fallen petals in warm water, filter (1.5); acids give pink, bases give green or dull colour (1.5).	1½+1½
17	Acid + base give salt and water (1); lime water added drop by drop to lemon juice with litmus (1); colour change shows the nature changes (1).	1+1+1
	OR: Antacid for indigestion (1); lime in acidic soil (1); baking soda on ant bite (1).	1+1+1
18	Keep chopped onion with cloth strips overnight (1); the smell is present in acids but lost in bases (2).	1+2
	OR: Bee sting is acidic, treated with a base like baking soda (1.5); wasp sting is basic, treated with vinegar (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Substance that changes colour with acids and bases (1). (ii) Paste, dip filter paper, dry and cut (2). (iii) Yellow in acid, red in base (2).	1+2+2
	OR: (i) An indicator (1). (ii) Reacts with a base or acid coated on paper (2). (iii) Turmeric and litmus (1). (iv) Write with soap solution, then spray an acid (1).	1+2+1+1
20	(i) Acid + base = salt + water (1). (ii) Lemon juice and lime water with indicator (2). (iii) Lime for soil; antacids for acidity (2).	1+2+2
	OR: (i) Excess acid (1). (ii) Antacid (1). (iii) It is acidic (1). (iv) Bile and pancreatic juice are basic (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Exploring Substances: Acidic, Basic, and Neutral** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** red	acids change blue litmus to red
2	**(c)** neutral	no change in either litmus
3	**(b)** lactic acid	formed by souring
4	**(b)** reddish-brown	soap solution turns it red
5	**(b)** salt and water	acid + base
6	**(b)** organic matter (compost)	gives acids on decay
7	**(a)** disappears	olfactory indicator
8	**(c)** basic	helps remove grease
9	**(d)** neutral	both papers stay the same
10	**(d)** Distilled water	pure water

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The sting has formic acid (1); baking soda is a base and neutralises it (1).	1+1
12	They may be acidic, basic or poisonous and harm us (2).	2
13	Acetic, lactic and citric acid (2).	2
14	It turns reddish-brown (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Changes soil acidity (1.5); lowers fertility and harms plants (1.5).	1½+1½
16	Stomach juice has acid (1); bile and pancreatic juice are basic and neutralise acid (2).	1+2
17	Acid: sour, red litmus unchanged/blue to red (1.5); base: bitter, soapy, red litmus to blue (1.5).	1½+1½
	OR: P and S acidic (1); Q basic (1); R neutral (1).	1+1+1
18	It neutralises acids formed by bacteria after eating (3).	3
	OR: Turmeric does not change colour with acids (1.5); it only shows bases (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Colour changes show nature (2). (ii) Acid, base, neutral (2). (iii) To avoid wrong conclusion (1).	2+2+1
	OR: (i) Poor crop growth (1). (ii) Lime for acidic; compost for basic (2). (iii) Balances pH (2).	1+2+2
20	(i) Acidic waste (1). (ii) Litmus (1). (iii) Add lime (1). (iv) Balances acidity (2).	1+1+1+2
	OR: (i) Lemon, vinegar, curd, tamarind, amla (2). (ii) Sour (1). (iii) Some can be harmful (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.